COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S2
DATE PERFORMED:	10/07/2026	**DATE SUBMITTED:**	15/07/2026
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ACTIVITY NO. 1.1 [BASIC C++ PROGRAMMING]

1. PROCEDURE OUTPUT

ILO A

Sections	Answer
Header File Declaration Section	`#include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	`//Function that will take two numbers and display the sum`
	`int sumOfTwoNumbers(int fNum, int sNum);`
	`//Function that will return that inequality of the two numbers`
	`bool isGreater(int A, int B);`
	`//Function that returns two Boolean values and display the result of all logical operations then return true if it was a success.`
	`bool comparingBools(bool Num1, bool Num2);`

Main Function	`int main () {`
	
	`int number1, number2;`
	
	`std::cout << "Enter first`

```
number: " << std::endl;
    std::cin >> number1;

    std::cout << "Enter second
number: " << std::endl;
    std::cin >> number2;


    std::cout << "The sum of the
two numbers is: " <<
sumOfTwoNumbers(number1, number2)
<< std::endl;
    std::cout << "is the first
number greater than the second
number? " << (isGreater(number1,
number2)? "Yes" : "No") <<
std::endl;

    bool boolResult =
comparingBools(true, false);
    std::cout << "Was the
comparison successful? " <<
(boolResult ? "Yes" : "No") <<
std::endl;

    return 0;
}
```

Method Definition

```
int sumOfTwoNumbers(int fNum, int
sNum) {

    int sum = fNum + sNum;
    return sum;
}

bool isGreater(int A, int B) {
```

	``
	return A > B;
	}
	
	bool comparingBools(bool Num1, bool Num2) {
	    std::cout << "AND: " << (Num1 && Num2) << std::endl;
	    std::cout << "OR: " << (Num1 || Num2) << std::endl;
	    std::cout << "NOT A: " << (!Num1) << std::endl;
	    std::cout << "NOT B: " << (!Num2) << std::endl;
	    return true;
	}
	``

ILO B

Header File Section Declaration	`#include <iostream>`
Global Declaration Section	

Class Declaration and Method Definition Section	```
	class Triangle{
	    private:
	    double totalAngle, angleA, angleB, angleC;
	    public:
	    Triangle(double A, double B, double C);
	    void setAngles(double A, double B, double C);
	    const bool validateTriangle();
	};
	``

Main Function	`int main(){`

```
    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
    std::cout << "The shape is a
valid triangle.\n";
}
else {
    std::cout << "The shape is NOT
a valid triangle.\n";
}
return 0;
}
```

Method Definition

```
Triangle::Triangle(double A,
double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
void Triangle::setAngles(double A,
double B, double C) {
    angleA = A;
    angleB = B;
    angleC = C;
    totalAngle = A+B+C;
}
const bool
Triangle::validateTriangle() {
    return (totalAngle <= 180);
}
```

```
PS C:\Users\Benedict\Downloads\Hands-on
The shape is a valid triangle.
```

This is the output of ILO B and it outputs that the shape is a valid triangle because the values add up to a 180 which is the proper angle of the triangle. I noticed that the private classes initialized are all of the angles because they are constant and shouldn't be changed outside of their respective classes. While the public classes are the

ones where you'll be initializing its value.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

ILO C_1

```cpp
#include <iostream>

int main() {
    int Var1, Var2;
    int Swapped;

    std::cout << "Please Enter Your First Variable: ";
    std::cin >> Var1;

    std::cout << "Please Enter Your Second Variable: ";
    std::cin >> Var2;

    std::cout << "Your Original Variables are: " << std::endl;
    std::cout << "First Variable:  " << Var1 << std::endl;
    std::cout << "Second Variable: " << Var2 << std::endl;

    Swapped = Var1;
    Var1 = Var2;
    Var2 = Swapped;


    std::cout << "Your Swapped Variables are now: " << std::endl;
    std::cout << "First Variable: " << Var1 << std::endl;
    std::cout << "Second Variable: " << Var2 << std::endl;
    return 0;
}
```

ILO C_2

```cpp
#include <iostream>


float toFahrenheit(float Kelvin);

int main() {
    float Kelvin;

    std::cout << "--- Kelvin to Fahrenheit---" << std::endl;

    std::cout << "Please Enter Temperature in Kelvins (*K): ";
    std::cin >> Kelvin;

    std::cout << "Your temperature in From Kelvin to Fahrenheit is: " <<
toFahrenheit(Kelvin) << std::endl;
}

float toFahrenheit(float Kelvin) {
    return ((Kelvin - 273.15)*9/5)+ 32;
}
```

ILO C_3

```cpp
#include <iostream>
#include <cmath>

float distanceTwoPoints(float x1, float x2, float y1, float y2);

int main() {
    float x1, x2, y1, y2;

    std::cout << "Please Enter your X Variables: " << std::endl;
    std::cout << "x1: ";
    std::cin >> x1;
    std::cout << "x2: ";
    std::cin >> x2;

    std::cout << "Please Enter your Y Variables: " << std::endl;
    std::cout << "y1: ";
    std::cin >> y1;
    std::cout << "y2: ";
    std::cin >> y2;

    float DistancebetweenTwoPoints = distanceTwoPoints(x1, x2, y1, y2);
    std::cout << "The distance between two points is: " <<
DistancebetweenTwoPoints<< std::endl;
}

float distanceTwoPoints(float x1, float x2, float y1, float y2) {
    float dx, dy;

    dx = x2 - x1;
    dy = y2 - y1;

    return sqrt(dx * dx + dy * dy);
}
```

ILO C_4

```
#include <iostream>
#include <cmath>

class Triangle{
    private:
    double totalAngle, AngleA, AngleB, AngleC;
    double Height, Base, totalArea;
    double SideA, SideB, SideC, PerimeterTotal;

    public:
    Triangle(double A, double B, double C);
    double Area(double D, double E);
    double Perimeters(double F, double G, double H);
    void setAngles(double A, double B, double C);
    void setArea(double D, double E);
    void setPerimeters(double F, double G, double H);
    void triangleClassification();
    const bool validateTriangle();

};

Triangle::Triangle(double A, double B, double C) {
    AngleA = A;
    AngleB = B;
    AngleC = C;
    totalAngle = A + B + C;

}

double Triangle::Area(double D, double E) {
    Base = D;
    Height = E;

    totalArea = 0.5 * Base * Height;
```

```
    return totalArea;
}

double Triangle::Perimeters(double F, double G, double H) {
    SideA = F;
    SideB = G;
    SideC = H;

    PerimeterTotal = F + G + H;
    return PerimeterTotal;
}

void Triangle::setAngles(double A, double B, double C) {
    AngleA = A;
    AngleB = B;
    AngleC = C;
    totalAngle = A+B+C;
}

void Triangle::setArea(double D, double E) {
    Base = D;
    Height = E;

    totalArea = 0.5 * Base * Height;
}
void Triangle::setPerimeters(double F, double G, double H) {
    SideA = F;
    SideB = G;
    SideC = H;

    PerimeterTotal = F + G + H;
}

void Triangle::triangleClassification() {
    if (AngleA, AngleB, AngleC < 90) {
        std::cout << "This Triangle is Acute-Angled" << std::endl;
    } else if (AngleA, AngleB, AngleC == 90) {
        std::cout << "This is Triangle is Classified as 'Others'." <<
```

```
std::endl;
    } else {
        std::cout << "This Triangle is Obtused-Angled." << std::endl;
    }
}

const bool Triangle::validateTriangle() {
    return (totalAngle <= 180);

}

int main(){
    double totalAngle;

    Triangle set1(40, 30, 110);
    if(set1.validateTriangle()){
        std::cout << "The shape is a valid triangle.\n";
    } else {
        std::cout << "The shape is NOT a valid triangle.\n";
}
    std::cout << "Therefore, ";
    set1.triangleClassification();

    double area = set1.Area(10, 6);
    std::cout << "The Area of the Triangle is: " << area << std::endl;

    double perimeter = set1.Perimeters(25, 15, 20);
    std::cout << "The Perimeter of a Triangle is: " << perimeter <<
std::endl;


return 0;
}
```

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

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Tools Used: **Claude AI**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **Code assistance, explanation of concepts**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **Use of AI was utilized for guidance regarding syntaxes, information about the format of the code and how it'll work, recalling concepts like classes.**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

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Conclusion:

The lessons I learned from this activity include the declaration of functions and classes, as well as how initializing them before the main code makes debugging much easier whenever errors occur. In ILO A, the task mainly required the use of input, output, functions, and basic syntax to compare two variables. However, the biggest challenge I encountered was managing the Boolean syntax correctly, since I could not clearly remember how to use Boolean values and conditions properly.

For ILO B, the code was already provided, but I struggled with understanding how classes work and why certain variables need to be declared as public or private. As for the first three supplementary activities, they were relatively simple because they only required the use of functions along with basic input and output syntax. However, the fourth supplementary activity took me the longest to complete because of the many concepts I had to keep track of, such as working with classes and determining what type of triangle was being represented. I even had to do additional research on the different types of triangles and their properties to complete the task correctly.

Overall, I believe my C++ skills are still lacking in several areas. I need to improve my ability to remember the proper syntax and structure of C++ programs, as well as develop a deeper understanding of object-oriented programming concepts, especially classes. I hope to continue

studying these topics so that I can apply them more effectively in my future programming projects.